

- Q.19 Explain in brief the Bulk polymerization technique. (CO2)
- Q.20 Describe the role of Stabilizers in plastic. (CO4)
- Q.21 Explain the structure, two properties and two uses of polypropylene. (CO3)
- Q.22 Describe the Suspension polymerization technique for polymer production. (CO2)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Write a note on the following term :
 a) Viscosity Average molecular weight (CO1)
 b) Solution polymerization (Co2)
- Q.24 Explain structure, properties and uses of Urea formaldehyde resin. (CO3)
- Q.25 Discuss Blow molding processing technique with neat and labelled sketch. (CO4)

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6th Sem/ Chemical Subject : Polymer Technology

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 What is the term for the process of forming a polymer from Monomers? (CO1)
 a) Polymerization b) De-Polymerization
 c) Macromolecule d) Oligomers
- Q.2 The example of natural polymer is _____. (CO1)
 a) PVC b) Rubber
 c) Polypropylene d) Polystyrenes
- Q.3 Bakelite is a/an _____ polymer (CO1)
 a) Thermoplastic b) Thermosetting
 c) Elastomers d) Fibers
- Q.4 Which of the following monomeric unit is used for the production of polyvinyl chloride? (CO3)
 a) Vinyl chloride b) Ethylene
 c) Ethyne d) Ethane

Q.5 What is the function of Antioxidant stabilizer in plastic? (CO4)

- a) To improve impact resistance
- b) To enhance thermal stability
- c) To prevent degradation from oxygen attack
- d) To increase flexibility

Q.6 What is the formula for calculation of number Average molecular weight ($\bar{M}_n$)? (CO1)

- a) $\frac{n_i m_i}{n_i}$
- b) $\frac{n_i m_i}{\sum n_i}$
- c) $\frac{\sum n_i m_i}{n_i}$
- d) $\frac{\sum n_i m_i}{\sum n_i}$

Where n_i , m_i are number of molecule and molecular weight of i^{th} polymer respectively.

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 The Melting point of a polymer _____ as its Average Molecular weight increases. (CO1)

Q.8 Bulk and solution poly merization are Homogenous polymerization. (True/False) (CO2)

Q.9 The order of chain growth polymerization Reaction in initiation $\rightarrow$ Propagation $\rightarrow$ Termination. (True/False) (CO2)

Q.10 Write the monomer used for Teflon. (CO3)

Q.11 Define degree of polymerization. (CO1)

Q.12 Give any one example of fiber. (CO1)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Describe the linear and branched chain polymer with examples. (CO1)

Q.14 Explain the following terms : (CO1)

- a) Number Average molecular weight
- b) Weight Average molecular weight

Q.15 Explain condensation polymerization with example. (CO2)

Q.16 Define Initiator with one example and its role in polymerization. (CO2)

Q.17 Write down the structure, two properties and two uses of Nylon 66. (CO3)

Q.18 Draw neat and labelled sketch of Injection molding machine. (CO4)